cretion stream or re-processing in the photosphere of the M giant star (Jablonski & Pereira 1997).

5.8. *Galactic Orbit*

We retrieve the right ascension, declination, and proper motion of IGR J16194-2810 from *Gaia* DR3 (see Table 2). Combining these initial conditions with our best-fit parallax and center-of-mass RV, and assuming the Milky Way gravitational potential described by `MilkyWayPotential2022` in `gala` (Price-Whelan 2017), we use `galpy` (Bovy 2015) to integrate the galactic orbit of IGR J16194-2810 back in time by 1 Gyr in the left and middle panels of Figure 9.

We find that the galactic orbit of IGR J16194-2810 is puffier and less circular than the solar orbit, but still confined within ± 1 kpc of the disk midplane. This puffiness is possibly due to the natal kick (see Section 6.1).

Under the assumption that IGR J16194-2810 participates in the Galactic rotation, Ratti et al. (2010) derive the peculiar velocity of the system as a function of its center-of-mass velocity (which was unknown at the time). They find an erroneously high minimum peculiar velocity of 280 ± 66 km s^{-1}. This, in turn, follows from an inaccurate assumption of a high source proper motion of 1.3 ± 4.7 mas yr^{-1} in RA and -20.2 ± 4.7 mas yr^{-1} in Dec. from UCAC2 — now superseded by the values provided in *Gaia* DR3 (see Table 2). Based on their large derived minimum peculiar velocity, Ratti et al. (2010) postulate that the binary is either a halo object or received a natal kick. We have shown that the former is unlikely, and that the latter is not to be expected in the case of a SyXB.

We plot a Toomre diagram of stars in the vicinity of IGR J16194-2810 in the right panel of Figure 9. We choose stars within a circle of radius 1° centered on the target coordinates that have zeropoint-corrected parallaxes within 0.1 mas of the system and RVs from *Gaia* DR3 with uncertainties < 5 km s^{-1}. We find that the space velocity of IGR J16194-2810 is consistent with that of nearby stars.

6. DISCUSSION

6.1. *Formation History*

As stated in the introduction, SyXBs present a challenge for binary evolution models because they need to survive common-envelope evolution without undergoing a merger event and emerge as a stripped helium (He) star + main-sequence star binary in a relatively wide orbit. Furthermore, the supernova (SN) of the He star will (1) result in significant mass loss and (2) impart a natal kick to the NS that, when taken together, are likely to unbind the binary (Hills 1983; Brandt & Podsiadlowski 1995; Hobbs et al. 2005). Here, we probe the combination of mass loss, kicks, and initial periods that could reproduce the orbit of IGR J16194-2810. We simulate 10^7 random combinations of He star masses, initial orbital periods, natal kicks, and kick orientations, and retain the ones that result in a bound orbit at least as wide as observed today. In doing so, we sample from uniform pre-SN distributions of $M_{\rm He} \in [2\ M_\odot, 5\ M_\odot]$, $P_{\rm orb} \in [0.1\ {\rm d}, 400\ {\rm d}]$, and $v_{\rm kick} \in [0\ {\rm km\ s^{-1}}, 500\ {\rm km\ s^{-1}}]$, respectively. We assume a neutron star mass of $1.4\ M_\odot$ and a companion mass of $1\ M_\odot$. Informed by our analysis of the galactic orbit of IGR J16194-2810 in Section 5.8, we only retain the combinations of initial parameters that result in post-SN relative center-of-mass velocities ≤ 50 km s^{-1}. We do not place a constraint on eccentricity because the orbit was likely circularized by tides when the companion star became a red giant, after the supernova.

We provide the distributions of the allowed He star masses, initial orbital periods, natal kicks, and kick orientations, along with the resulting eccentricities and relative systemic velocities, in Figure 10. As expected, we find that a bound orbit wider than $P = 193$ days favors smaller He star masses (i.e. smaller amounts of mass loss), larger initial orbital periods, and smaller natal kicks. The 1σ upper limits on the allowed He star mass and kick velocity are $3.87\ M_\odot$ and 44.6 km s^{-1}, respectively. We also find that reproducing the observed orbit requires the kick to be either roughly aligned or anti-aligned with the motion of the He star. Peak probabilities occur at $\theta = 120^\circ$ or 60°, respectively (due to rotational symmetry). Following the supernova, we expect the resulting orbit to be eccentric with a small relative systemic velocity. However, as mentioned above, tidal interactions will eventually circularize the final orbit.

We plot the probability that the orbit of IGR J16194-2810 remains bound after a supernova as a function of natal kick and He star mass in Figure 11. We again follow the prescription of Brandt & Podsiadlowski (1995), assuming an initial orbital period of 100 days, a red giant mass of $1\ M_\odot$, and a neutron star remnant mass of $1.4\ M_\odot$. At each kick velocity and/or He star mass, we simulate 1000 random kick orientations and compute the probability that the resulting NS + low-mass star binary remains bound. As expected, the bound probability decreases with increasing He star mass (due to more extensive mass loss) and with increasing kick velocity.

We simulate the range of natal kicks that allow the final orbit to both be bound and at least as wide as observed as a function of initial orbital period in the top panel of Figure 12. We repeat the analysis in Figure 11, except that we assume a fixed He star mass of $3\ M_\odot$, allow the initial orbital period to vary, and additionally require the final orbit to have $P > 193$ days. We keep track of the minimum and maximum natal kicks that allow for final orbits that satisfy these conditions as a function of initial orbital period. The median allowed kick velocity decreases with increasing initial orbital period, with the allowed range becoming more stringent as well. In the bottom panel of Figure 12, we plot the probability that the final orbit is bound and at least as wide as observed as a function of initial orbital period for the median allowed natal kick. Specifically, we simulate 1000 random orientations at each median natal kick and report the corresponding probability as a function of initial orbital period. We find that the probability increases as the initial orbital period increases, approaching ~ 0.16 as $P_{\rm init}$ reaches ~ 400 days. These simulations suggest that — given typical NS kick velocities of ~ 300 km s^{-1} — a majority of would-be SyXBs become

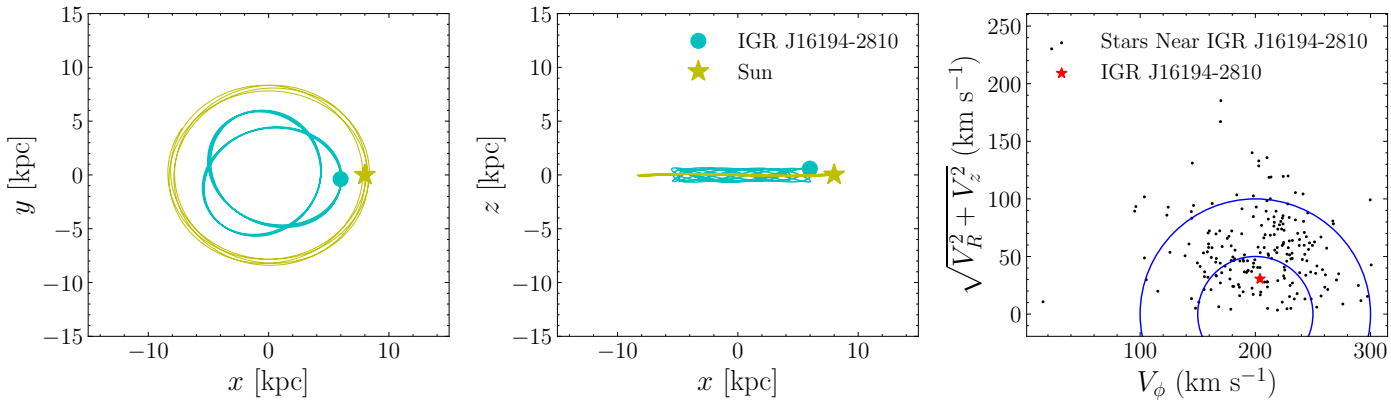

Figure 9. Left and Middle Panels: Galactic orbit of IGR J16194-2810 integrated back in time by 1 Gyr based on its current location and spatial velocity. The orbit is clearly puffier and less circular than the solar orbit, shown for reference. This indicates that IGR J16194-2810 was originally orbiting in the plane of the Galactic disk before acquiring a larger spatial velocity due to the supernova of the NS progenitor. Right Panel: Toomre diagram comparing velocities in the direction of the Galactic rotation to velocities perpendicular to the Galactic rotation for stars in the vicinity of IGR J16194-2810. The blue semi-circles are centered on the average rotational speed of $\langle V_\phi \rangle = 200$ km s^{-1} and have radii of 50 km s^{-1} and 100 km s^{-1}. We find that the space velocity of IGR J16194-2810 is consistent with that of nearby stars.

unbound during the supernova, and only systems born with relatively weak kicks survive.

6.2. *Future Binary Evolution*

Based on our derived orbital parameters, we calculate the possible future evolution of the IGR J16194-2810 binary using the Modules for Experiments in Stellar Astrophysics (MESA; Paxton et al. 2011, 2013, 2015, 2018, 2019; Jermyn et al. 2023). We start with a 1 $M_\odot$ star orbiting a 1.4 $M_\odot$ neutron star (treated as a point mass) in a 200 day orbit. We use the mass transfer prescription of Kolb & Ritter (1990) and the mass transfer efficiency formalism described by Soberman et al. (1997). We assume fully non-conservative mass-transfer (i.e. $\beta = 1$).

We plot the evolution of the donor star on a Hertzsprung-Russell diagram, focusing on the main sequence and red giant branch, in the top left panel of Figure 13. We find that the red giant in IGR J16194-2810 has not yet achieved Roche lobe overflow (i.e. it is currently experiencing mass loss via winds). During the Roche lobe overflow phase, which lasts $\sim$ 4 Myr, the red giant experiences a period of heightened mass transfer. This phase is marked by a change in slope of the red giant branch on the H-R diagram. We show the evolution of the mass loss rate as a function of age and orbital period in the bottom left and bottom right panels of Figure 13, respectively. We find that, on average, the red giant experiences a mass loss of $\sim 2 \times 10^{-7}$ $M_\odot$ yr^{-1} during Roche lobe overflow. In total, the donor loses around $\sim$ 0.6 $M_\odot$ over the course of Roche lobe overflow (top right panel of Figure 13). Eventually, the red giant loses its envelope and descends down the white dwarf (WD) cooling track. The binary ends up with a slowly cooling $\sim$ 0.4 $M_\odot$ helium WD orbiting a 1.4 $M_\odot$ NS in a $\sim$ 900 day orbit, comparable to several of the longest-period WD + millisecond pulsar binaries known (e.g. Tauris 2011).

An upper limit for the accretion rate of the NS today (assuming $v_{\rm wind} \gg c_s$) is given by the Bondi-Hoyle-Lyttleton (BHL) equation:

$$\dot{M}_{\rm BHL} = \frac{G^2 M_{\rm NS}^2 \dot{M}_{\rm wind}}{v_{\rm wind}^4 a^2} \tag{8}$$

where a is the separation between the NS and its red giant companion. Approximating $v_{\rm wind}$ as the photospheric escape velocity of the red giant (about 84 km s^{-1} assuming our best-fit parameters) and adopting the present-day mass loss rate from our MESA simulation (about 4.94×10^{-9} $M_\odot$ yr^{-1}) for $\dot{M}_{\rm wind}$, we find that the predicted average accretion rate today is approximately 1.6×10^{-10} $M_\odot$ yr^{-1}. Assuming an accretion efficiency of $\eta = 0.1$, the corresponding accretion luminosity (produced mainly in X-rays) is $L_X = \eta \dot{M}_{\rm BHL} c^2 \approx 10^{36}$ erg s^{-1}. At a distance of about 2.1 kpc, this implies an observed X-ray flux of $F_X \approx 2 \times 10^{-9}$ erg s^{-1} cm^{-2}. Our predicted accretion luminosity is about two orders of magnitude higher than the values reported in Table 2. This could be due to a reduced accretion rate, reduced accretion efficiency, or both. Indeed, Yungelson et al. (2019) point out that, for slow NS rotation, the NS's magnetic field causes the gravitationally captured plasma to form a hot, convective shell above the magnetosphere. This plasma enters the magnetosphere at a steady rate regulated by plasma cooling (due to the Rayleigh-Taylor instability), sub-sonically settling down with reduced accretion rate $\dot{M} \lesssim 0.5\, \dot{M}_{\rm BHL}$.

6.3. *Comparison of SyXBs to the Population of Wide NS + MS Binaries*

Astrometric orbital solutions from *Gaia* DR3 have enabled the discovery of 21 NS candidates orbiting low-mass stars in wide orbits (El-Badry et al. 2024a,b). Based on our derived results, we can compare the space density and implied birth rate of SyXBs to that of the population of wide NS +

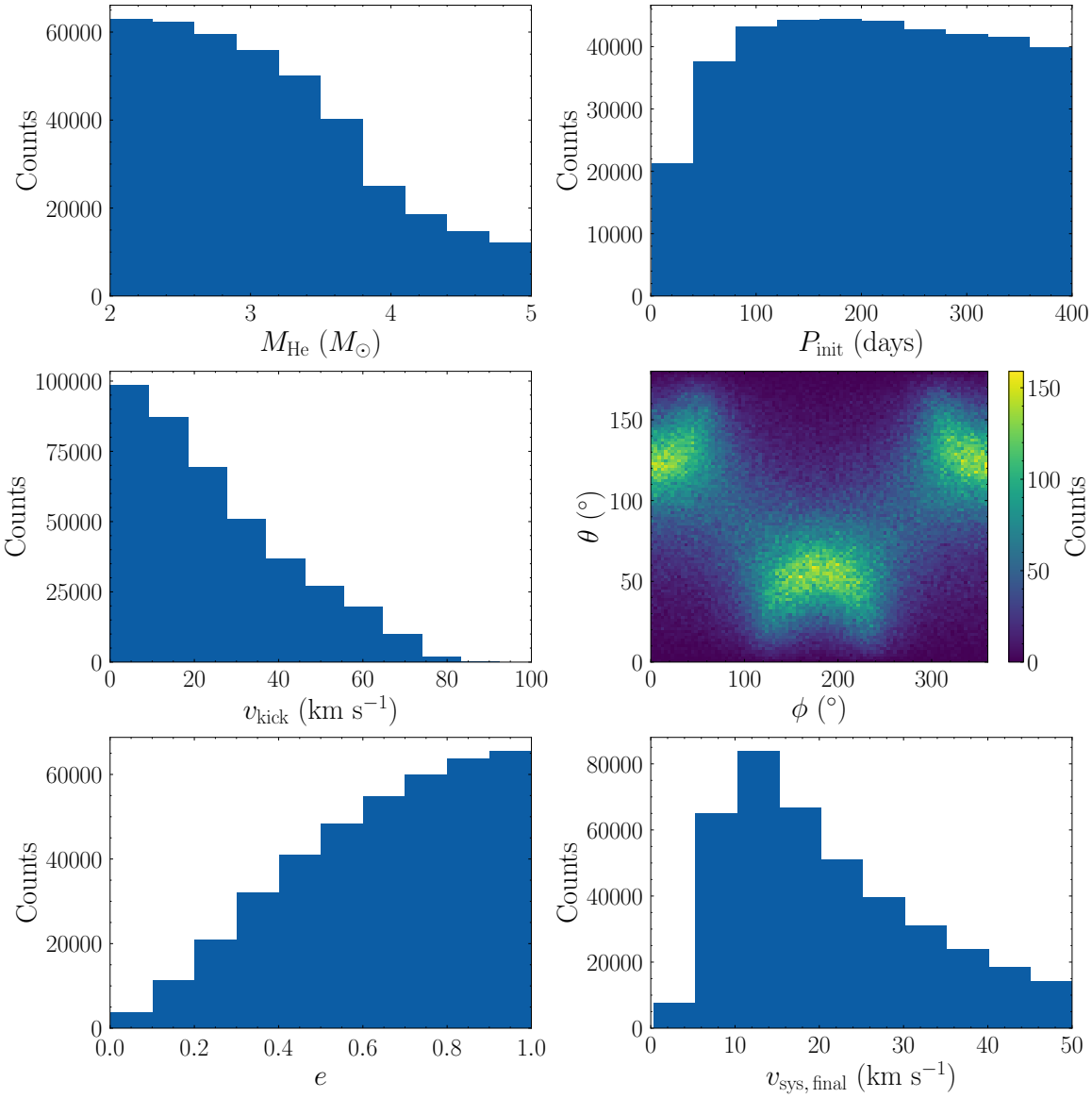

Figure 10. Distributions of parameters that allow the final orbit of IGR J16194-2810 to both be bound and as least as wide as observed, assuming $M_{\rm NS} = 1.4\ M_\odot$ and $M_2 = 1\ M_\odot$. Top Row: Distributions of He star masses and initial orbital periods. Middle Row: Distributions of natal kicks and natal kick orientations. Bottom Rows: Distributions of final orbital eccentricities and relative systemic velocities. We find that bound NS + red giant binaries favor small He star masses, large initial orbital periods, and small natal kicks roughly aligned or anti-aligned with the motion of the He star. Prior to tidal circularization, the resulting orbit is expected to be eccentric, with a small relative systemic velocity.

MS binaries. At a distance of about 2.1 kpc, IGR J16194-2810 is probably the nearest SyXB with $P_{\rm orb} \lesssim 1000$ days (i.e. detectable in *Gaia* DR3). This suggests a space density ~ 100 times lower than what is estimated for the *Gaia* NS + MS binaries, the closest of which is ~ 250 pc away from the Sun (El-Badry et al. 2024b). However, the X-ray bright lifetime of SyXBs is likely only on the order of ~ 10 Myr (see Section 6.2). Interpolating on MIST stellar models with `isochrones`, we compute the middle 68% of the main sequence lifetimes of the companions of the NS candidates reported by El-Badry et al. (2024b) to be approximately 6–9 Gyr: almost 1000 times longer than the X-ray bright lifetime of SyXBs.

Taken together, these space densities and lifetimes suggest a comparable or higher birth rate for SyXBs relative to the *Gaia* NS + MS binary population. Given that the *Gaia* sample is not yet complete, it appears likely that most SyXBs are descendants of the wide NS + MS binaries being revealed by *Gaia*.

A caveat to this analysis is that we are probably still missing some NS + MS binaries. The upcoming *Gaia* DR4 is expected to lead to more discoveries and shed further light on this population (El-Badry 2024). In addition, the SyXB